

Rainfall Intelligence

Data Cleaning and Exploratory Analysis of Tamil Nadu Rainfall

Project Documentation — Sprint I

Project Objective

This project analyzes Tamil Nadu rainfall telemetry data to convert raw rainfall observations into a cleaner, structured, and analysis-ready dataset. The first sprint focuses on data cleaning and exploratory data analysis (EDA), with the goal of understanding rainfall behavior across time, districts, monitoring stations, extreme events, and data coverage.

1. Real-World Problem

Rainfall is not uniform across Tamil Nadu. It changes by location, season, time period, and rainfall event intensity. Organizations working with agriculture, water resources, flood monitoring, drought assessment, and climate-related planning need reliable rainfall information before making decisions.

However, raw telemetry data is difficult to use directly. It may contain inconsistent data types, missing values, duplicate records, timestamp formatting issues, unnecessary location or hierarchy columns, and a large number of observations that are difficult to interpret without systematic analysis.

Problem Statement

The project addresses the problem of transforming raw Tamil Nadu rainfall telemetry records into a clean and understandable dataset and identifying meaningful rainfall patterns that can support further analysis and future prediction work.

2. Who Has This Problem?

- Agriculture and irrigation teams — to understand rainfall availability and seasonal patterns.
- Water resource management teams — to study rainfall distribution and support water planning.
- Flood and disaster-management teams — to identify locations and periods associated with heavy rainfall events.
- Climate and environmental analysts — to explore rainfall trends and regional variation.
- Data analysts and data scientists — to obtain a reliable dataset for dashboards, statistical analysis, and future machine-learning models.

3. What This Project Tries to Find

The analysis is designed to answer questions such as:

- What is the overall distribution of hourly rainfall measurements?
- How does average rainfall vary from year to year?
- What is the monthly rainfall pattern?
- Which districts have higher average rainfall in the available data?
- Which monitoring stations show higher average rainfall?
- How frequently do high-rainfall events occur?

- In which years do rainfall events of at least 100 mm occur more frequently?
- Which stations have the strongest data coverage?
- How much data is available for each year and station?

4. Proposed Solution

The solution is a Python-based rainfall data analysis pipeline. The raw dataset is loaded with Pandas, checked for quality problems, converted into suitable datatypes, enriched with time-based columns, analyzed using grouped summaries, and visualized using Matplotlib.

The result of Sprint I is an analysis-ready rainfall dataset and a set of exploratory findings that can become the foundation for later dashboards or machine-learning prediction work.

5. Technology Stack

Technology	Purpose
Python	Main programming language
Pandas	Data loading, cleaning, transformation and grouping
NumPy	Numerical/data-analysis support
Matplotlib	Charts and visual analysis
Jupyter Notebook	Interactive project development and documentation
CSV	Dataset storage and exchange

6. Data Cleaning Process

The notebook performs the following data-quality checks and transformations:

- Inspect the dataset shape to understand the number of records and columns.
- View sample records using the first rows of the dataset.
- Inspect column information and datatypes.
- Generate descriptive statistics for numerical fields.
- Check missing values using null-value counts.
- Check duplicate records.
- Convert Data Acquisition Time into a proper datetime datatype.
- Convert Telemetry Hourly Rainfall (mm) into a numeric datatype.
- Determine the available start date and end date.
- Identify the number of monitoring stations and districts.
- Create Year, Month, Month_Name and Date fields for time-based analysis.
- Remove unnecessary administrative/geographical hierarchy columns from the analysis dataset.
- Export the resulting EDA dataset for further use.

7. Exploratory Data Analysis

Overall Rainfall Distribution

A histogram is used to understand how hourly rainfall measurements are distributed, including the concentration of lower values and the presence of high-rainfall observations.

Year-wise Analysis

The project calculates record count, average rainfall, median rainfall, and maximum rainfall for each year. A bar chart is used to compare average rainfall between years.

Monthly Analysis

Monthly record counts, averages, medians, and maximum values are calculated. A line chart shows the monthly rainfall pattern across the year.

District-level Analysis

Rainfall is grouped by District to compare record count, average, median, and maximum rainfall. The top districts by average rainfall are visualized.

Station-level Analysis

The same type of analysis is performed for individual monitoring stations. This helps identify stations with higher average rainfall and compare spatial variation.

Extreme Rainfall Analysis

The notebook specifically identifies observations at or above 100 mm and 1000 mm, lists the highest rainfall observations, and counts ≥ 100 mm events by year.

Data Coverage Analysis

The project checks how many records are available for each station and each year. It also calculates unique Station-Day combinations to understand the breadth of monitoring coverage.

8. What Problem Is Actually Solved?

The project does not claim to predict future rainfall in Sprint I. Instead, it solves the earlier and necessary data-analysis problem: **making raw rainfall telemetry data usable and extracting the patterns required for informed analysis.**

- Raw rainfall data → structured and checked dataset.
- Unclear timestamps → standardized datetime information.
- Rainfall values stored as text/inconsistent types → numeric rainfall measurements.
- Large raw dataset → summarized information by year, month, district, and station.
- Hidden heavy-rainfall events → explicit extreme-event analysis.
- Unknown monitoring coverage → station and yearly coverage analysis.
- Unnecessary hierarchy fields → reduced analysis dataset for later work.

9. Project Uniqueness

The project is not limited to a basic dataset cleaning exercise. It combines data quality assessment with rainfall-specific exploratory questions. The analysis connects temporal behavior, geographic/station-level variation, extreme rainfall events, and monitoring coverage in one workflow.

This makes the project useful as a foundation for a future rainfall intelligence system, where the cleaned data can later be used for dashboards, anomaly detection, forecasting, or machine-learning models.

10. Expected Practical Value

- Provides a cleaner foundation for rainfall analysis.
- Makes rainfall patterns easier to compare across time and locations.
- Highlights heavy-rainfall observations that may deserve further investigation.
- Shows where monitoring data is strong or limited.
- Creates a reusable foundation for future analytical or ML stages.

11. Current Scope and Limitation

This documentation describes Sprint I based on the supplied notebook. The notebook performs cleaning and EDA; it does not yet implement a rainfall prediction model, deployment system, or real-time alerting system. Those can be treated as future development stages.

Important: Numerical final findings such as the exact average rainfall, maximum rainfall, top district, and number of extreme events should be filled from the actual dataset execution. They are intentionally not invented in this report.

12. Future Enhancement

- Build an interactive Power BI or web dashboard.
- Add stronger anomaly and outlier analysis.
- Develop rainfall forecasting/prediction models using the cleaned dataset.
- Evaluate machine-learning models with appropriate metrics.
- Create location-based rainfall alerts for heavy-rainfall events.

- Automate ingestion of new telemetry records for continuous analysis.

13. Project Conclusion

Rainfall Intelligence — Sprint I establishes a reliable analytical foundation for Tamil Nadu rainfall data. By combining systematic data cleaning with exploratory analysis, the project transforms raw telemetry observations into meaningful summaries about rainfall distribution, temporal patterns, geographic variation, extreme events, and data coverage. The resulting dataset and analysis workflow can be extended into visualization, forecasting, and intelligent rainfall-monitoring applications.